Test Series No.

1.14

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 24/03/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

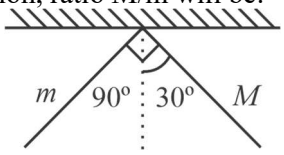
Student's Signature: _____ **Invigilator's Signature:** _____

1. A spherical body is spinning around its central axis without any external torque. Due to rise in temperature its volume increases by 3%. Then percentage change in its angular speed is:

(1) –2% (2) –1%

(3) –3% (4) 1%

2. Two uniform rods of equal lengths but different masses (*m* and *M*) are rigidly joined to form an L–shaped body which is then pivoted as shown. If in equilibrium, the body is in the shown configuration, ratio M/m will be:



(1) 2 (2) $\frac{1}{\sqrt{3}}$

(3) $\sqrt{2}$ (4) $\sqrt{3}$

3. A point charge q = 50 μC is located in the *x*–*y* plane at the point with position vector $\vec{r}_0 = (2\hat{i} + 3\hat{j})$ m. What is the magnitude of electric field at the point with position vector $\vec{r} = (8\hat{i} - 5\hat{j})$ m?

(1) 1200 V/m (2) 4×10^{-2} V/m

(3) 900 V/m (4) 4500 V/m

4. A sphere of radius *a* is uniformly charged throughout its volume with a volume charge density ρ. The electric field at a distance *r* from the centre of the sphere ($r < a$) is: (ϵ_0 is permittivity of free space)

(1) $\frac{\rho r}{\epsilon_0}$ (2) $\frac{\rho r}{3\epsilon_0}$

(3) $\frac{\rho}{2\epsilon_0}$ (4) $\frac{\rho r}{4\epsilon_0}$

5. At what distance on the axis from the centre of a circular coil is the magnetic field half of that at the centre? Given $2^{\frac{2}{3}} = 1.587$ (Here *R* is radius of the circular coil)

(1) 0. 566 *R* (2) $\frac{R}{\sqrt{2}}$

(3) 0.766 *R* (4) $\frac{3R}{2}$

6. A Carnot engine is made to work between 200°C and 0°C first and then between 0°C and –200°C. The ratio of efficiencies $\left(\frac{\eta_1}{\eta_2}\right)$ of the engine in the two cases is:

(1) $\frac{473}{273}$ (2) 1 : 1

(3) 1 : 2 (4) $\frac{273}{473}$

7. A process is carried on a fixed amount of an ideal gas in such a way that the rate of collision of molecules per unit area of vessel walls remains constant. Then:

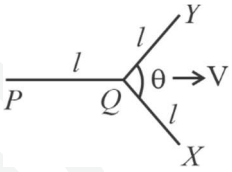
(1) $P \propto V$

(2) $V \propto T$

(3) Process is isothermal

(4) Process is adiabatic

8. A conducting wire in the shape of Y, with each side of length *l*, is moving in a uniform magnetic field *B* perpendicular to the plane with a uniform speed *V* as shown in the figure. The induced emf between the two ends X and Y of the wire will be:



(1) zero

(2) $2B\ell V$

(3) $2B\ell V \sin(\theta/2)$

(4) $2B\ell V \cos(\theta/2)$

9. The quantity L/RCV has the units of:
(where *L* is self-inductance, *R* is resistance, *C* is capacitance and *V* is voltage)

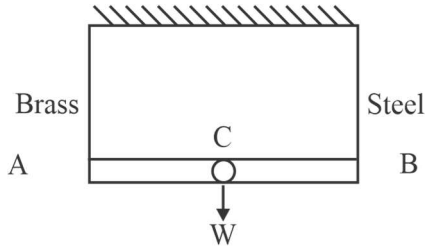
(1) charge

(2) $(\text{charge})^{-1}$

(3) current

(4) $(\text{current})^{-1}$

10. A 2 m long light metal rod AB is suspended from the ceiling horizontally by means of two vertical wires of equal length tied to its ends. One wire is of brass and has cross–section of 0.2 x 10^{-4} m² and the other is of steel with 0.1×10^{-4} m² cross–section. In order to have equal stresses in the two wires, a weight is hung from the rod. The position of the weight along the rod from end A should be:



(1) 66.6 cm

(2) 133 cm

(3) 44.4 cm

(4) 155.6 cm

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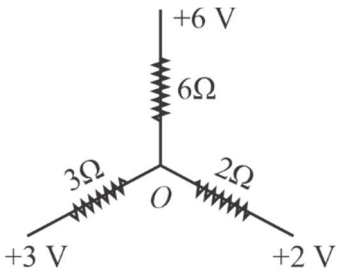
11. The speed v of a particle moving along a straight line, when it is at a distance x from a fixed point on the line is given by: $v = \sqrt{108 - 9x^2}$. Then, (All quantities are in SI units.)
(1) The motion is uniformly accelerated.
(2) The magnitude of acceleration at a distance 3 m from the fixed point is 27 m/s².
(3) The magnitude of acceleration at a distance 3 m from the fixed point is 9 m/s².
(4) The maximum displacement of the particle from the fixed point is 4 m.

12. A uniform rope of mass M = 0.1 kg and length L = 10 m hangs freely from the ceiling. A transverse pulse is produced at lower end. then, (g = 10 m/s²)
(1) Speed of the transverse pulse increases linearly from bottom to the top.
(2) Speed decreases linearly from bottom to the top.
(3) Speed remains constant along the rope.
(4) Time taken by the transverse pulse to travel full length is 2 sec.

13. In a Young's double slit interference experiment, the fringe pattern is observed on a screen placed at a distance D. The slits are separated by d and are illuminated by light of wavelength λ. The distance from the central point, where the intensity falls to half the maximum is:
(1) $\frac{\lambda D}{3d}$ (2) $\frac{\lambda D}{2d}$
(3) $\frac{\lambda D}{d}$ (4) $\frac{\lambda D}{4d}$

14. When an object is placed at a distance of 25 cm from a mirror, the magnification is m_1 . The object is moved 15 cm away from mirror with respect to the earlier position, magnification becomes m_2 . If $\frac{m_1}{m_2} = 4$, the focal length of the mirror is:
(1) –10 cm.
(2) –30 cm.
(3) –15 cm.
(4) –20 cm.

15. In the figure shown, what is the potential of junction O?



(1) 3 V (2) zero
(3) –6 V (4) 1 V

16. A convex lens forms a real image on a screen placed at a distance of 60 cm from the object. When the lens is shifted towards the screen by 20 cm, another image of the object is formed on the screen. The focal length of the lens is:
(1) 45 cm. (2) 40/3 cm.
(3) 30 cm. (4) 12 cm.

17. A wave represented by $y = a\cos(kx - \omega t)$, is superimposed with another wave to produce a stationary wave, such that x = 0 is a node. The equation of the other wave is:
(1) $y = a\sin(kx + \omega t)$
(2) $y = -a\cos(kx - \omega t)$
(3) $y = -a\cos(kx + \omega t)$
(4) $y = -a\sin(kx - \omega t)$

18. The de Broglie wavelength of a neutron of mass 1.6 × 10^{-24} gm moving with an energy of 0.04 eV is:
(1) 0.143 Å. (2) 1.43 Å.
(3) 0.72 Å. (4) 0.82 Å.

19. The binding energy per nucleon for deutron ($_1H^2$) and helium ($_2He^4$) are 1.1 MeV and 7.0 MeV respectively. The energy released when two deutrons fuse to form a helium nucleus ($_2He^4$) is:
(1) 28 MeV.
(2) 2.2 MeV.
(3) 23.6 MeV.
(4) 18.6 MeV.

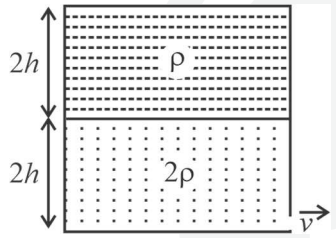
20. The ratio of the magnitude of magnetic dipole moment to the angular momentum is a universal constant for hydrogen like atomic ions. The value of this universal constant is:
(1) $\frac{e^2}{8\epsilon_0 m}$ (2) $\frac{e}{2m}$
(3) $\frac{\pi m}{2e}$ (4) $\frac{m^2}{2e\epsilon_0}$

21. A diver looking up through the water in a lake sees the outside world contained in a circular horizon. If the refractive index of water is 4/3 and the eye of the diver is situated 12 cm below the surface, the radius of the horizontal circle will be:
(1) $12 \times 3 \times \sqrt{5}$ (2) $12 \times \frac{4}{3}$
(3) $12 \times \frac{3}{\sqrt{7}}$ (4) $12 \times \sqrt{7} \times 3$

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22. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The speed of the planet is smallest in its elliptical path when it is farthest from the sun.
Reason R: In rotational motion of the planets around the sun, the law of conservation of angular momentum is obeyed.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

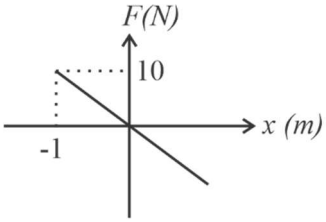
23. The velocity (v) of the liquid coming out of a small hole of a large vessel containing two different liquids of densities 2ρ and ρ as shown in figure is:

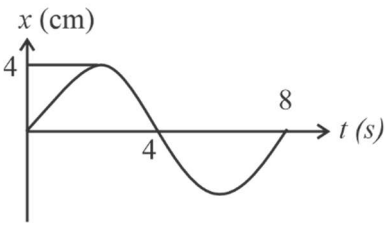


(1) $\sqrt{6gh}$
(2) $2\sqrt{gh}$
(3) $2\sqrt{2gh}$
(4) $\sqrt{gh}$

24. The amount of work done in forming a soap bubble of radius r = 1 cm (surface tension T = 3×10^{-2} N/m) is:
(1) 37.38 μJ (2) 40.20 μJ
(3) 75.36 μJ (4) 20.10 μJ

25. F–x and x–t graph of a particle in SHM are as shown in figure. Match the following:

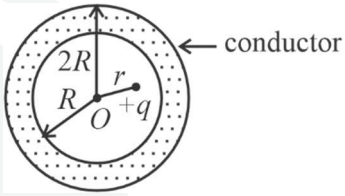




List-I		List-II	
(A)	Mass of the particle	(I)	$\frac{\pi}{4}$ SI unit
(B)	Maximum kinetic energy of particle	(II)	$\left(\frac{160}{\pi^2}\right)$ SI unit
(C)	Angular frequency of particle	(III)	(8.0×10^{-3}) SI unit

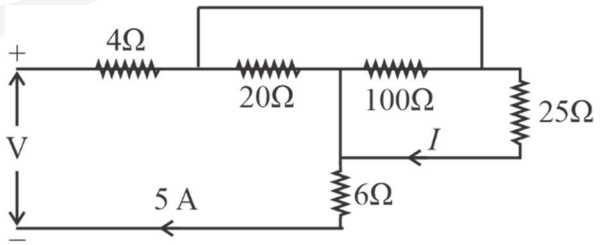
(1) (A)–(II), (B)–(III), (C)–(I)
(2) (A)–(III), (B)–(II), (C)–(I)
(3) (A)–(III), (B)–(I), (C)–(II)
(4) (A)–(II), (B)–(I), (C)–(III)

26. A point charge q is placed at a distance r from the centre O of an uncharged conducting spherical shell of inner radius R and outer radius $2R$. The electric potential at the centre of the shell will be:



(1) $\frac{q}{4\pi\epsilon_0}\left(\frac{1}{r}-\frac{1}{2R}\right)$
(2) $\frac{q}{4\pi\epsilon_0 r}$
(3) $\frac{q}{4\pi\epsilon_0}\left(\frac{1}{r}+\frac{1}{2R}\right)$
(4) $\frac{q}{4\pi\epsilon_0}\left(\frac{1}{r}-\frac{1}{R}\right)$

27. In the circuit shown the potential difference V must be:

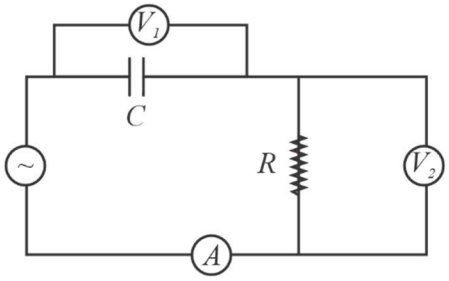


(1) 50 V (2) 100 V
(3) 75 V (4) 25 V

28. A ray of light passes through an equilateral glass prism in such a manner that the angle of incidence is equal to the angle of emergence and each of these angles is equal to $\frac{3}{4}$ of the angle of prism. The angle of minimum deviation is:
(1) 45°. (2) 70°.
(3) 39°. (4) 30°.

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29. The diagram shows a capacitor C and a resistor R connected in series to an AC source. V_1 and V_2 are voltmeters and A is an ammeter.



Consider the following statements:

I. Readings in A and V_2 are always in phase.

II. Reading in V_1 is ahead in phase with reading in V_2.

III. Readings in A and V_1 are always in phase.

Correct statements are:

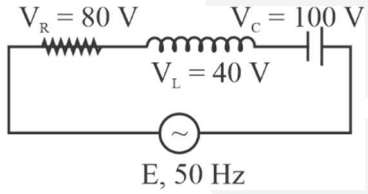
(1) I only

(2) II only

(3) I and II only

(4) II and III only

30. The rms value of alternating emf E in the given circuit will be:



(1) 100 V

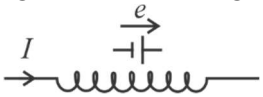
(2) 20 V

(3) 220 V

(4) 140 V

31. Given below are two statements:

Statement I: Figure shows an emf e induced in a coil due to rightward decreasing current.



Statement II: In the coil self induced emf is $e = -L\left(\frac{di}{dt}\right)$.

In the light of the above statements, choose the most appropriate answer from the options given below:

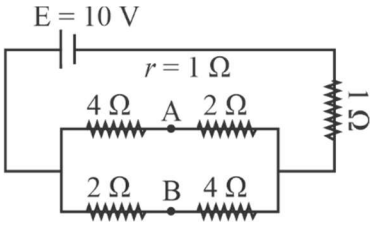
(1) Both Statement I and Statement II are correct.

(2) Both Statement I and Statement II are incorrect.

(3) Statement I is correct but Statement II is incorrect.

(4) Statement I is incorrect but Statement II is correct.

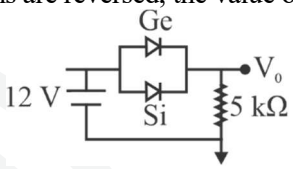
32. In the circuit shown below, the cell has an e.m.f. of 10 V and internal resistance of 1 ohm. Other resistances are shown in the figure. The potential difference $V_A - V_B$ is:



(1) 6 V (2) 4 V

(3) 2 V (4) –2 V

33. Ge and Si diodes conduct at 0.3 V and 0.7 V respectively. In the given circuit, if Ge diode connections are reversed, the value of V_0 changes by:



(1) 0.2 V (2) 0.4 V

(3) 0.6 V (4) 0.8 V

34. The escape velocity from the surface of a planet is v_e. If the mass of the planet becomes 4 times and its radius becomes half, the new escape velocity will be:

(1) v_e (2) $2v_e$

(3) $2\sqrt{2}v_e$ (4) $v_e/2$

35. In a semiconducting material the mobilities of electrons and holes are μ_e and μ_h respectively. Which of the following is **true**?

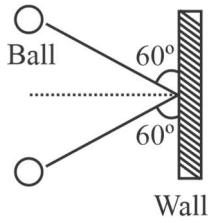
(1) $\mu_e > \mu_h$

(2) $\mu_e < \mu_h$

(3) $\mu_e = \mu_h$

(4) $\mu_e < 0; \mu_h > 0$

36. A ball of mass 3 kg, moving with a speed of 100 m/s, strikes a wall at an angle 60° (as shown in figure). The ball rebounds at the same speed and remains in contact with the wall for 0.2 sec; the force exerted by the ball on the wall is:



(1) $1500\sqrt{3}$ N

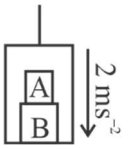
(2) 1500 N

(3) $300\sqrt{3}$ N

(4) 300 N

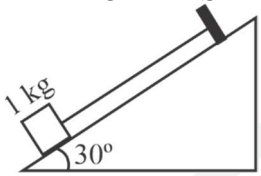
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37. The elevator shown in figure is descending, with an acceleration of 2 m s^{-2}. The mass of the block A = 0.5 kg. The force exerted by block A on block B is: (g = 10 m s^{-2})



(1) 2 N (2) 4 N
(3) 6 N (4) 8 N

38. In the given incline system, the coefficient of friction between two surfaces is μ = 0.8. The tension in the string shown in the figure is: (g = 10 m/s^2)

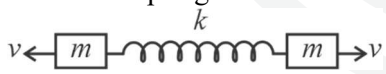


(1) 0 N (2) 6 N
(3) 4 N (4) 8 N

39. Force acting on a particle moving in a straight line varies with velocity (v) of the particle as $F = K / v$, where K is a constant. The work done by this force in time t is:

(1) $\frac{K}{v^2} \cdot t$ (2) $2Kt$
(3) Kt (4) $\frac{2Kt}{v^2}$

40. Two blocks each of mass m are connected to a spring of spring constant k. If both are given velocity v in opposite directions, then the maximum elongation of the spring is:



(1) $\sqrt{mv^2 / k}$ (2) $\sqrt{2mv^2 / k}$
(3) $\sqrt{mv^2 / 2k}$ (4) $2\sqrt{mv^2 / k}$

41. A ball of mass m approaches a wall of mass M ($>> m$) with speed 4 m/s along the normal to the wall. The speed of wall is 1 m/s towards the ball. The speed of the ball after an elastic collision with the wall is:
(1) 5 m/s away from the wall
(2) 9 m/s away from the wall
(3) 3 m/s away from the wall
(4) 6 m/s away from the wall

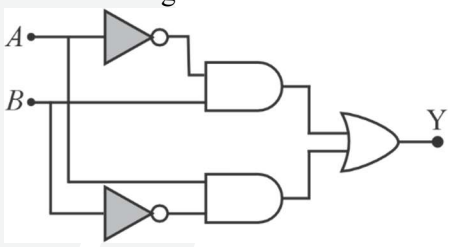
42. The length and breadth of a rectangular sheet are measured as 5.20 cm and 3.0 cm respectively. The area of the sheet with correct significant figures is:
(1) 15.6 cm^2 (2) 15.60 cm^2
(3) 16 cm^2 (4) 15.600 cm^2

43. If $Z = \frac{A^2 B}{C^3}$, where A, B and C are measured quantities, then the maximum relative error in Z in terms of relative errors of A, B and C, is:

(1) $2\left(\frac{\Delta A}{A}\right)+\left(\frac{\Delta B}{B}\right)+3\left(\frac{\Delta C}{C}\right)$
(2) $2\left(\frac{\Delta A}{A}\right)+\left(\frac{\Delta B}{B}\right)-3\left(\frac{\Delta C}{C}\right)$
(3) $\left(\frac{\Delta A}{A}\right)+\left(\frac{\Delta B}{B}\right)+\left(\frac{\Delta C}{C}\right)$
(4) $2\left(\frac{\Delta A}{A}\right)-\left(\frac{\Delta B}{B}\right)+3\left(\frac{\Delta C}{C}\right)$

44. A satellite is orbiting very close to the surface of Earth. If the radius of Earth is R and acceleration due to gravity is g, the orbital speed of the satellite is:
(1) $\sqrt{gR}$ (2) $\sqrt{2gR}$
(3) $\sqrt{gR/2}$ (4) gR

45. The truth table for the following logic circuit is: (Refer to the image for the four truth table options)



(1)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(2)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(3)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	1

(4)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

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46. The Henry's law constants K_H for three gases A, B, C in water are 160 kbar, 4.0×10^{-5} kbar and 40 kbar, respectively. The solubility of these gases in water at same T and P follows the order:

(1) A > C > B
(2) B > C > A
(3) C > A> B
(4) C > B > A

47. The highest number of helium atoms is present in:

(1) 3 mol of helium
(2) 6 g of helium
(3) 8 u of helium
(4) 11.2 L of helium at STP

48. Identify the major product C in the following reaction sequence:

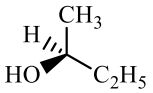
$$CH_3-CH_2-CH_2-CH_2-Br \xrightarrow{NaCN} A \xrightarrow[\text{partial hydrolysis}]{OH^-} B \xrightarrow{Br_2/NaOH} C(\text{major})$$

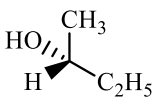
(1) propylamine
(2) butylamine
(3) pentanamide
(4) α-bromopentanoic acid

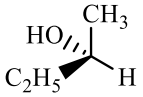
49. Which one of the following statements is **wrong**?

(1) The uncertainty principle can be written as $\Delta x \Delta p \geq \frac{h}{4\pi}$.
(2) For hydrogen-like species, all orbitals with the same principal quantum number n have the same energy.
(3) In a hydrogen atom, the energy of the 3s orbital is less than the energy of the 3p orbital.
(4) de Broglie wavelength is $\lambda = \frac{h}{mv}$, where v is the particle's velocity.

50. Find out relation in above structures:

(I) 

(II) 

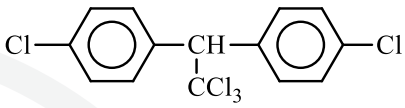
(III) 

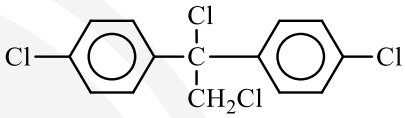
(1) I and II are identical
(2) I and III are enantiomer
(3) II and III are identical
(4) II and III are enantiomer

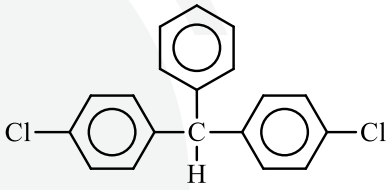
51. Three solutions are prepared at 300 K by dissolving 9 g of glucose $(C_6H_{12}O_6)$ in 300 mL of water (P_1), 9 g of urea (CH_4N_2O) in 300 mL of water (P_2), and 9 g of sucrose ($C_{12}H_{22}O_{11}$) in 300 mL of water (P_3). The decreasing order of osmotic pressure is:

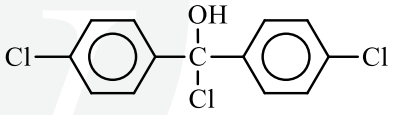
(1) $P_1 > P_2 > P_3$
(2) $P_2 > P_1 > P_3$
(3) $P_3 > P_1 > P_2$
(4) $P_2 > P_3 > P_1$

52. Trichloroacetaldehyde, CCl_3CHO reacts with chlorobenzene in presence of sulphuric acid and produces:

(1) 

(2) 

(3) 

(4) 

53. The compound C_7H_8 undergoes the following reactions:

$$C_7H_8 \xrightarrow{3Cl_2/\Delta} A \xrightarrow{Br_2/Fe} B \xrightarrow{Zn/HCl} C$$

The product 'C' is:

(1) 3-Bromo-2,4,6-trichlorotoluene
(2) o-bromotoluene
(3) m-bromotoluene
(4) p-bromotoluene

54. Values of work function (Wo) for a few metals are given below:

Metal	Li	Na	K	Mg	Cu
W_0(eV)	2.30	2.50	2.10	3.70	4.70

The number of metals which will show photoelectric effect when light of wavelength 500 nm falls on it is:
(Given: h = 6.6×10^{-34} Js, c = 3×10^8 ms^{-1}, e = 1.6×10^{-19} C)

(1) 1
(2) 2
(3) 3
(4) 4

[Contd...

55. Which of the following statements are **correct**?

A. The process of adding an electron to a neutral gaseous atom is always exothermic.

B. The process of removing an electron from an isolated gaseous atom is always endothermic.

C. The first ionization energy of Al is less than that of Mg.

D. The electronegativity of nitrogen is the same in NH_3 and NF_3.

E. Potassium is the most electropositive among group 1 elements.

Choose the **correct** answer from die options given below:

(1) B and C only

(2) B, C and D only

(3) A, B and D only

(4) B, D and E only

56. What mass of 80% pure $CaCO_3$ is required to completely neutralise 40 mL of 0.50 M HNO_3 according to the reaction:

$$CaCO_3(s) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + CO_2(g) + H_2O(\ell)$$

(1) 1.00 g

(2) 1.10 g

(3) 1.25 g

(4) 2.50 g

57. The **correct** statement regarding the comparison of staggered and eclipsed conformations of ethane is:

(1) The staggered conformation is less stable than eclipsed because it has torsional strain.

(2) The eclipsed conformation is more stable than staggered because it has no torsional strain.

(3) The eclipsed conformation is more stable than staggered even though eclipsed has torsional strain.

(4) The staggered conformation is more stable than eclipsed because staggered has minimum torsional strain.

58. Consider the following relations for emf of an electrochemical cell:

A. E°_{cell} = (Oxidation potential of anode) – (Reduction potential of cathode)

B. E°_{cell} = (Oxidation potential of anode) + (Reduction potential of cathode)

C. E°_{cell} = (Reduction potential of anode) + (Reduction potential of cathode)

D. E°_{cell} = (Oxidation potential of anode) – (Oxidation potential of cathode)

Which of the above relations are **correct**?

(1) A and B

(2) C and D

(3) B and D

(4) C and A

59. For a reaction:

$$2X(g) + Y(g) \rightarrow Z(g) + W(g) - q$$

If ΔS for the reaction is negative, which statement is **correct** about spontaneity?

(1) Possible only at low temperature

(2) Possible only at high temperature

(3) Never possible

(4) Possible at all temperatures

60. Consider the statements regarding reducing sugars and common disaccharides:

A. All monosaccharides (aldose or ketose) are reducing sugars.

B. A sugar is reducing if it can reduce Fehling's solution and Tollens' reagent.

C. Sucrose is reducing because it yields glucose and fructose on hydrolysis.

D. Maltose is non-reducing sugar.

Choose the **correct** option:

(1) A, B and D only

(2) A and B only

(3) B and C only

(4) A, B and C only

61. The **correct** statement regarding an electrophile is:

(1) It is always negatively charged and accepts electrons from another electrophile

(2) It is generally neutral and accepts electrons from another electrophile

(3) It may be neutral or positively charged and accepts an electron pair from a nucleophile

(4) It is always negatively charged and accepts an electron pair from a nucleophile

62. For an isolated system, which option is **correct**?

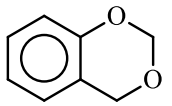
(1) $q = 0$ and $w = 0$

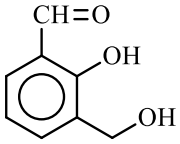
(2) $q \neq 0$ and $w = 0$

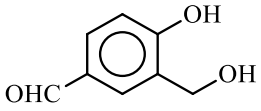
(3) $q = 0$ and $w \neq 0$

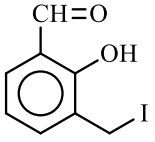
(4) $q \neq 0$ and $w \neq 0$

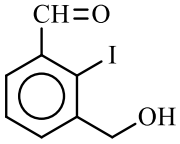
[Contd...

63.  $\xrightarrow[\text{(ii) } CHCl_3, NaOH]{\text{(i) HI}}$ product (major):

(1) 

(2) 

(3) 

(4) 

64. Calculate the pH of 0.10 M salt of a weak monobasic acid and weak monoacidic base at 25°C. (Given $pK_a = 4.6$, $pK_b = 5.4$).

(1) 7.4 (2) 6.6

(3) 8.2 (4) 7.0

65. Consider liquid-vapour equilibrium: Liquid $\rightleftharpoons$ Vapour. Which relation is **correct**?

(1) $\frac{d\ln P}{dT} = \frac{\Delta H_{vap}}{RT}$ (2) $\frac{d\ln P}{dT} = -\frac{\Delta H_{vap}}{R}$

(3) $\frac{d\ln P}{dT^2} = \frac{\Delta H_{vap}}{T^2}$ (4) $\frac{d\ln P}{dT} = \frac{\Delta H_{vap}}{RT^2}$

66. Which thermodynamic quantity (as written) is **not** a state function?

(1) Enthalpy

(2) Temperature

(3) Pressure

(4) w under isothermal condition

67. Match the **List-I** with **List-II.**

List-I		List-II	
(A)	XeF_4	(I)	Pyramidal
(B)	XeF_6	(II)	Square planar
(C)	$XeOF_4$	(III)	Distorted octahedral
(D)	XeO_3	(IV)	Square pyramidal

Choose the **correct** answer from the options given below;

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-III, C-IV, D-I

(3) A-II, B-III, C-I, D-IV

(4) A-III, B-IV, C-I, D-II

68. Which complex shows the shortest C–O bond length (free CO bond length = 1.13 Å)?

(1) $Ni(CO)_4$

(2) $[Co(CO)_4]^-$

(3) $[Fe(CO)_4]^{2-}$

(4) $[Mn(CO)_6]^+$

69. Which statement is **true** regarding hydrogen bonding?

(1) It is a directional interaction

(2) Its strength is comparable to a covalent O–H bond

(3) It requires no lone pair on the electronegative atom

(4) It occurs only between identical molecules

70. The addition of a catalyst during a chemical reaction alters which quantity?

(1) Entropy change of reaction

(2) Internal energy change of reaction

(3) Enthalpy change of reaction

(4) Activation energy of reaction

71. A first-order reaction has rate constant $k = 2\times10^{-2}\ s^{-1}$. Time required for reactant mass to reduce from 20 g to 5 g is:

(1) 17.3 s (2) 34.7 s

(3) 69.3 s (4) 138.6 s

72. Which complex is used as an anticancer agent?

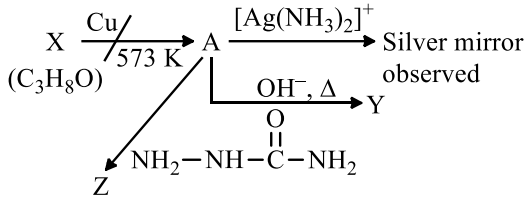
(1) fac-$[Co(NH_3)_3Cl_3]$

(2) cis-$[PtCl_2(NH_3)_2]$

(3) trans-$[PtCl_2(NH_3)_2]$

(4) mer-$[Co(NH_3)_3Cl_3]$

73. Consider the reactions:



Identify A, X, Y and Z

(1) A-Methoxyethane, X-Methanol, Y-Methanoic acid, Z-Semicarbazide.

(2) A-Propanal, X-Propanol, 2-Methylpent-2-enal, Z-Semicarbazone.

(3) A-Propanol, X-Propanal, Y-Butanone, Z-Hydrazone

(4) A-Methoxymethane, X-Ethanoic acid, Y-Acetate ion. Z-hydrazine

[Contd...

74. $CH_3CHO \xrightarrow{NaOBr} A + B$ (Alkyl halide)

$A \xrightarrow{LiAlH_4} C$. The final product C is:

(1) CH_3OH (2) CH_3CHO

(3) HCHO (4) CH_3Br

75. Select the **incorrect** statement for ligand behaviour.

(1) Neutral ligands can sometimes produce larger Δ_0 than anionic ligands

(2) Chelating ligands generally form more stable complexes than non-chelating ligands

(3) π-bonding ligands like CO are classical (purely σ-donor) ligands

(4) Donor ability depends on electronegativity of the donor atom

76. Based on typical first transition-series trends, consider:

A. The second ionisation enthalpy of Cu is unusually high because Cu^+ attains a stable $3d^{10}$ configuration.

B. The third ionisation enthalpy of Mn is unusually high because Mn^{2+} is $3d^5$ (half-filled) and resists further oxidation.

C. Across the series, M^{2+}/M standard electrode potentials are all positive and increase uniformly from Sc to Zn.

D. M^{3+}/M^{2+} electrode potential for Mn is strongly positive because Mn^{2+} is particularly stable.

Choose the **correct** option:

(1) A, B and D only (2) A and C only

(3) B and C only (4) A, B and C only

77. The ion that gives characteristic flame colour of brick red is:

(1) Na^+ (2) K^+

(3) Ca^{2+} (4) Ba^{2+}

78. An example of a disproportionation reaction is:

(1) $2KMnO_4 \rightarrow K_2MnO_4 + MnO_2 + O_2$

(2) $2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

(3) $2CuBr \rightarrow CuBr_2 + Cu$

(4) $2NaBr + Cl_2 \rightarrow 2NaCl + Br_2$

79. Which of the following molecules has the maximum dipole moment?

(1) HF (2) CH_4

(3) NH_3 (4) NF_3

80. Which of the following cations gives a blood-red colour with potassium thiocyanate (KSCN)?

(1) Fe^{3+} (2) Cu^{2+}

(3) Co^{2+} (4) Ni^{2+}

81. Which of the following is an **incorrect** statement?

(1) $PbCl_4$ is covalent in nature

(2) $SiCl_4$ is easily hydrolysed

(3) GeX_4 (X = F, Cl, Br, I) is more stable than GeX_2

(4) $SnCl_4$ is ionic in nature

82. Number of ambidentate ligands among the following is _____:

$NO_2^-, NCS^-, CN^-, NH_3, C_2O_4^{2-}, NO_3^-$

(1) 2

(2) 3

(3) 4

(4) 5

83. Which of the following statements is **false**?

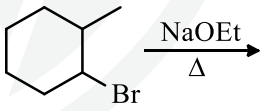
(1) Carius method is employed for the estimation of nitrogen in an organic compound.

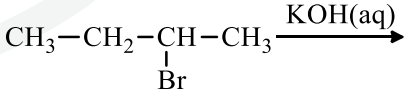
(2) A sealed carius tube is used in estimating sulphur or halogens by oxidation with fuming HNO_3.

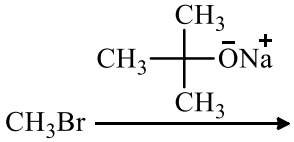
(3) Kjeldahl's method is used for estimation of nitrogen in organic compounds.

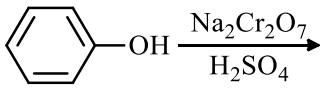
(4) On oxidising organic phosphorus to phosphoric acid, it may be precipitated as $MgNH_4PO_4$ and on ignition as $Mg_2P_2O_7$ using magnesia mixture.

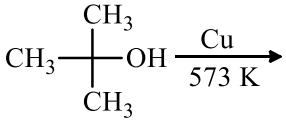
84. The reactions which can be applied to prepare an alcohol or ether are:

A. 

B. 

C. 

D. 

E. 

Choose the **correct** answer from the options given below:

(1) B and E Only

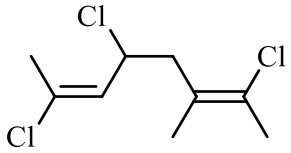
(2) B and C Only

(3) A, C and D Only

(4) B and D Only

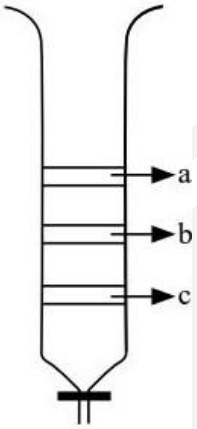
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85. Total number of stereoisomers possible for the given structure is:



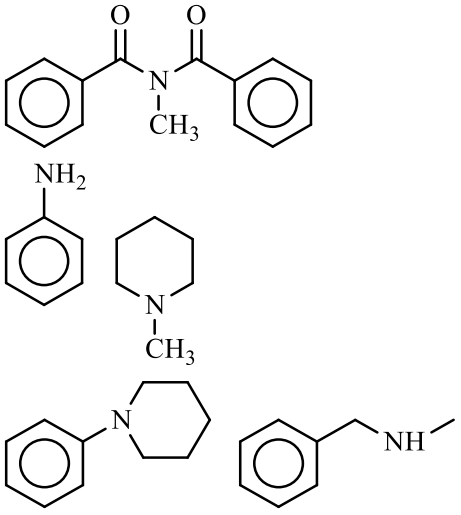
(1) 4
(2) 6
(3) 8
(4) 2

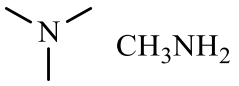
86. From the figure of column chromatography given below, identify **incorrect** statement.



(A) Compound 'c' is less polar than 'a' and 'b'
(B) Compound 'a' is least polar
(C) Compound 'b' comes out of the column before 'c' and after 'a'
(D) Compound 'a' spends more time in the column
Choose the **correct** answer from the options given below;
(1) A, B and C only
(2) B, C and D only
(3) A, B and D only
(4) B and C only

87. Number of compounds which give reaction with Hinsberg's reagent is_____.





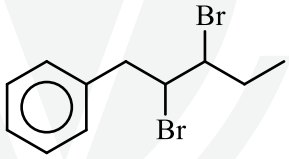
(1) 2
(2) 3
(3) 4
(4) 1

88. Match **List-I** with **List-II**.

List-I (Vitamin)		List-II (Deficiency disease)	
(A)	Vitamin A	(I)	Scurvy
(B)	Vitamin C	(II)	Increased blood clotting time
(C)	Vitamin D	(III)	Xerophthalmia
(D)	Vitamin K	(IV)	Rickets

Choose the **correct** answer from the option given below:
(1) A-III, B-I, C-IV, D-II
(2) A-II, B-IV, C-I, D-III
(3) A-IV, B-III, C-II, D-I
(4) A-I, B-II, C-III, D-IV

89. IUPAC name of given compound is:



(1) 3,4-dibromo-5-phenylpentane
(2) 1-(2,3-dibromo)pentylbenzene
(3) 2,3-dibromo-1-phenylpentane
(4) 2,3-dibromo-5-phenylpentane

90. Given:

$Co^{3+} + e^- \rightarrow Co^{2+};\ E^o = +1.81\,V$

$Pb^{4+} + 2e^- \rightarrow Pb^{2+};\ E^o = +1.67\,V$

$Ce^{4+} + e^- \rightarrow Ce^{3+};\ E^o + 1.61\,V$

$Bi^{3+} + 3e^- \rightarrow Bi;\ E^o = +0.20\,V$

Oxidizing power of the species will increase in the order:

(1) $Ce^{4+} < Pb^{4+} < Bi^{3+} < Co^{3+}$
(2) $Co^{3+} < Pb^{4+} < Ce^{4+} < Bi^{3+}$
(3) $Bi^{3+} < Ce^{4+} < Pb^{4+} < Co^{3+}$
(4) $Co^{3+} < Ce^{4+} < Bi^{3+} < Pb^{4+}$

[Contd...

91. The splitting of water is associated with:
(1) PS I located on the outer surface of the thylakoid membrane.
(2) PS II located on the inner side of the thylakoid membrane.
(3) both PS I and PS II.
(4) stroma lamellae.

92. All of the following belong to Order Polymoniales, **except**:
(1) *Petunia* (2) Mango
(3) Brinjal (4) *Datura*

93. Match **List-I** with **List-II**.

List-I (Organism)		**List-II (Defense mechanism)**	
(A)	Some frog species	(I)	Cardiac glycosides
(B)	Monarch butterfly	(II)	Thorns
(C)	Cactus	(III)	Poisonous/Distasteful
(D)	*Calotropis*	(IV)	Camouflage

Choose the **correct** answer from the options given below:
(1) A–IV, B–III, C–II, D–I
(2) A–III, B–IV, C–I, D–II
(3) A–IV, B–I, C–II, D–III
(4) A–III, B–II, C–IV, D–I

94. The middle lamella in a plant cell is a layer that holds different neighbouring cells together and is mainly composed of:
(1) calcium carbonate.
(2) cellulose.
(3) hemicellulose.
(4) calcium pectate.

95. Identify correct statements about honey bees.
A. Males have 32 chromosomes
B. Workers have 16 chromosomes
C. Males are produced through parthenogenesis
D. Male heterogamety is seen
E. Females produce gametes through meiosis
Choose the **correct** answer from the options given below:
(1) A, B and C only (2) B and E only
(3) C and E only (4) B and D only

96. The chief pigment associated with photosynthesis, appears _____ in the chromatogram.
(1) yellow-green (2) bright or blue-green
(3) yellow-orange (4) yellow

97. DNA as 'nuclein' was named by:
(1) Friedrich Meischer.
(2) Maurice Wilkins.
(3) Francis Crick.
(4) Rosalind Franklin.

98. More than one carpel that are free are found in all of the following, **except**:
(1) *Michelia* (2) *Papaver*
(3) Rose (4) Lotus

99. Presently, _____ per cent of all mammal species in the world face the threat of extinction.
(1) 12 (2) 23
(3) 32 (4) 31

100. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Decapitation is a widely used practice in tea plantations and hedge making.
Reason (R): Removal of shoot tips usually results in the removal of effect of cytokinin.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

101. Food material that is similar to amylopectin and glycogen is:
(1) floridean starch. (2) laminarin.
(3) mannitol. (4) pyrenoid.

102. Polygenic inheritance is seen in the inheritance of:
A. ABO blood group
B. Human height
C. Phenylketonuria
D. Flower colour in snapdragon
E. Human skin colour
Choose the **correct** answer from the options given below:
(1) B and C only (2) B and E only
(3) A and C only (4) C and D only

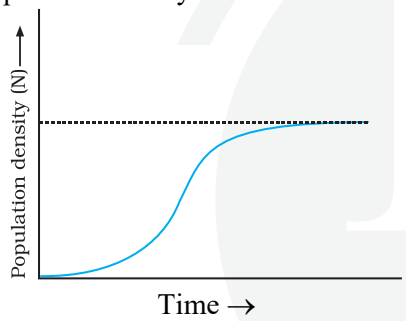
103. Meiosis is characterised by:
(1) one cycle of nuclear division and two cycles of DNA replication.
(2) two cycles of nuclear division and two cycles of DNA replication.
(3) two cycles of nuclear division and one cycle of DNA replication.
(4) one cycle of nuclear division and one cycle of DNA replication.

104. All of the following are nitrogen fixers and add organic matter to the soil and increase its fertility, **except**:
(1) *Methanobacterium*
(2) *Nostoc*
(3) *Anabaena*
(4) *Oscillatoria*

105. When fat acts as respiratory substrates in the respiratory pathway. R.Q value is:
(1) equals to 2 (2) equals to 1
(3) equals to 1.5 (4) less than 1

106. A typical angiosperm embryo sac, at maturity, has all of the following, **except**:
(1) Two synergids
(2) A central cell containing two polar nuclei
(3) A single egg cell
(4) Four cells forming the egg apparatus

107. Identify the **correct** statement regarding the graph of population density vs time.



(1) It shows exponential growth.
(2) It is shown by only some animals.
(3) It is known as the *Verhulst logistic growth curve*.
(4) It occurs under conditions of unlimited resources.

108. In mango and coconut:
(1) the middle mesocarp is edible.
(2) the fruit develops from a monocarpellary inferior ovary.
(3) the fruit is one-seeded.
(4) the mesocarp is fibrous.

109. The layers of the microsporangium wall that performs the function of protection and helps in dehiscence are:
(1) epidermis, endothecium, and tapetum.
(2) epidermis, middle layers, and tapetum.
(3) epidermis, endothecium, and middle layers.
(4) endothecium, middle layers, and tapetum.

110. In bacteria, which rRNA acts as a ribozyme and catalyses peptide bond formation?
(1) 18S rRNA (2) 23S rRNA
(3) 5.8S rRNA (4) 5srRNA

111. In a monocotyledonous stem, the tissue that is absent is:
(1) sclerenchymatous hypodermis.
(2) phloem parenchyma.
(3) parenchymatous ground tissue.
(4) water-containing cavities within vascular bundles.

112. Match **List-I** with **List-II.**

List-I (Structure w.r.t monocot embryo)		**List-II (Description)**	
(A)	Scutellum	(I)	Sheath enclosing the shoot apex
(B)	Coleoptile	(II)	Lateral cotyledon in grasses
(C)	Coleorrhiza	(III)	Sheath enclosing the radicle
(D)	Epiblast	(IV)	Rudimentary cotyledon

Choose the **correct** answer form the options given below:
(1) A–II, B–III, C–I, D–IV
(2) A–I, B–II, C–III, D–IV
(3) A–II, B–I, C–III, D–IV
(4) A–IV, B–I, C–III, D–II

113. Only gross superficial morphological characters were used for the classification in:
(1) natural classification system.
(2) artificial classification system.
(3) phylogenetic classification system.
(4) cytotaxonomy.

114. Baculoviruses are pathogens that attack:
(1) insects and other arthropods.
(2) brassicas.
(3) fruit trees.
(4) fishes.

115. Given below are two statements:
Statement I: The biosynthetic phase of photosynthesis is directly light driven and dependent of the presence of light.
Statement II: The enzymes required for the synthesis of sugars are localized in the thylakoid lumen of the chloroplast.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

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116. Match **List-I** with **List-II**.

List-I		List-II	
(A)	N-glycosidic linkage	(I)	Between amino acids
(B)	Phosphodiester bond	(II)	Nitrogenous base to sugar
(C)	Peptide bond	(III)	Between nitrogenous bases of opposite strands of DNA
(D)	Hydrogen bond	(IV)	Between two nucleotides

Choose the **correct** answer from the options given below:

(1) A–II, B–IV, C–I, D–III
(2) A–I, B–II, C–IV, D–III
(3) A–II, B–I, C–IV, D–III
(4) A–IV, B–II, C–I, D–III

117. Alternate phyllotaxy with pulvinate and stipulate leaves are seen in:

(1) pea.
(2) guava.
(3) sunflower.
(4) china rose.

118. Illegal introduction of which species for aquaculture is a threat to indigenous catfishes?

(1) *Lantana*
(2) *Eichhornia*
(3) *Clarias gariepinus*
(4) *Parthenium*

119. Match **List-I** with **List-II** regarding plastids:

List-I		List-II	
(A)	Elaioplasts	(I)	Store proteins
(B)	Amyloplasts	(II)	Store oils and fats
(C)	Aleuroplasts	(III)	Contain fat-soluble carotenoid pigments
(D)	Chromoplasts	(IV)	Store carbohydrates

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I
(2) A-II, B-IV, C-I, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-I, B-IV, C-II, D-III

120. Given below are two statements:

Statement I: Chromosomal aberrations are commonly observed in cancer cells.

Statement II: Alteration in chromosomes results in abnormalities because genes are located on chromosomes.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

121. In how many of the following karyogamy is followed by reductional division that produce spores endogenously?

A. *Claviceps*
B. *Ustilago*
C. *Puccinia*
D. *Alternaria*
E. *Neurospora*

(1) Two only (2) One only
(3) Three only (4) Four only

122. Formation of interfascicular cambium and cork cambium from fully differentiated parenchyma is:

(1) differentiation.
(2) dedifferentiation.
(3) redifferentiation.
(4) plasticity.

123. Identify the **correct** pair of product and source organism.

(1) Statin – *Propionibacterium sharmanii*
(2) Streptokinase – *Staphylococci* sp.
(3) Cyclosporin A – *Trichoderma polysporum*
(4) Acetic acid – *Aspergillus niger*

124. The adaxially placed palisade cells in a dorsiventral leaf is made up of:

(1) oval or round loosely arranged cells.
(2) elongated cells arranged vertically and parallel to each other.
(3) dead cells with highly thickened walls.
(4) colourless and empty cells.

125. Parasitism, a type of population interaction, is seen in:

(1) orchid on a mango branch.
(2) barnacles on a whale's back.
(3) *Cuscuta* growing on a hedge plant.
(4) Venus flytrap and insect.

126. In oocytes of some vertebrates _____ can last for months or years.
(1) zygotene
(2) pachytene
(3) diplotene
(4) diakinesis

127. Given below are two statements:
Statement I: Exons are non-coding sequences that do not appear in mature RNA.
Statement II: Split-gene arrangement represents an ancient feature of the genome.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

128. The formation of citric acid during the Krebs' cycle is a type of:
(1) decarboxylation reaction.
(2) phosphorylation reaction.
(3) condensation reaction.
(4) oxidation–reduction reaction.

129. Which of the following is **not** common between bryophytes and pteridophytes?
(1) Water required for fertilisation
(2) Multicellular sporophyte
(3) Presence of archegonia
(4) Independent sporophyte

130. Lactic Acid Bacteria:
(1) completely digest milk proteins.
(2) kills all stomach microbes.
(3) perform aerobic respiration during curd formation.
(4) increase the vitamin B_{12} content of curd.

131. Match **List-I** with **List-II.**

List-I (Chromosome)		**List-II (Genetic significance)**	
(A)	Chromosome 1	(I)	Location of gene for β-thalassemia
(B)	Chromosome 11	(II)	Location of genes for α-thalassemia
(C)	Chromosome 16	(III)	Chromosome with the highest number of genes
(D)	Chromosome 21	(IV)	Chromosome involved in Down's syndrome

Choose the **correct** answer form the options given below:
(1) A–III, B–I, C–II, D–IV
(2) A–II, B–III, C–I, D–IV
(3) A–III, B–II, C–I, D–IV
(4) A–I, B–II, C–III, D–IV

132. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Primary productivity varies in different types of ecosystems.
Reason (R): Primary productivity depends on varieties of environmental factors, availability of nutrients and photosynthetic capacity of plants.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

133. "Red tides" in the sea are often caused by the rapid multiplication of:
(1) *Euglena.*
(2) *Gonyaulax.*
(3) *Trypanosoma.*
(4) *Paramoecium.*

134. The genetic code is said to be "degenerate" because:
(1) one codon codes for multiple amino acids.
(2) more than one codon can code for the same amino acid.
(3) it is nearly universal.
(4) it has stop codons.

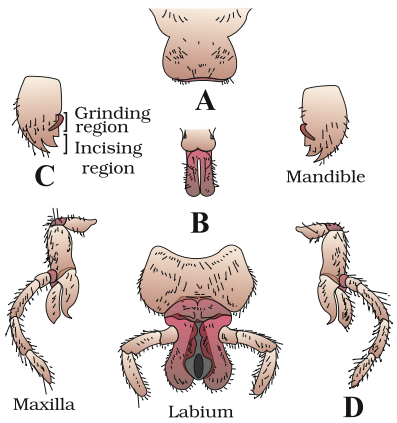
135. The hormone used to speed up the "malting process" in the brewing industry is:
(1) cytokinin.
(2) ethylene.
(3) GA_3.
(4) auxin.

136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The body of Aschelminthes is circular in cross-section, hence the name roundworms.
Reason R: Roundworms are bilaterally symmetrical, triploblastic, and true coelomate animals.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

137. How many of the following statements regarding epithelial tissues are **correct?**
A. The squamous epithelium is made of flattened cells with regular boundaries.
B. The cuboidal epithelium is composed of a single layer of cube-like cells.
C. The epithelium of the proximal convoluted tubule (PCT) of a nephron has microvilli.
D. Ciliated epithelium is mainly present in the inner surface of hollow organs.
E. Compound epithelium plays a major role in secretion and absorption.
(1) Two (2) Three
(3) Four (4) Five

138. In sponges, the pathway of water transport is helpful in all of the following, **except**:
(1) Food gathering
(2) Respiratory exchange
(3) Generation of nerve impulses
(4) Removal of waste

139. Carefully observe the following diagram of mouthparts in cockroach.



Which of the mouth parts acts as a tongue and lies within the cavity enclosed by the mouthparts?
(1) A
(2) B
(3) C
(4) D

140. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Coelenterata	(I)	Comb plates
(B)	Ctenophora	(II)	Water vascular system
(C)	Platyhelminthes	(III)	Flat body, suckers
(D)	Echinodermata	(IV)	Cnidoblasts

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-III, D-II
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-I, C-II, D-III
(4) A-II, B-III, C-IV, D-I

141. Which of the following cell junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells for rapid transfer of ions and small molecules?
(1) Tight junctions (2) Adhering junctions
(3) Gap junctions (4) Desmosomes

142. Select the combination of **correct** statements from the following regarding human endocrine system.
A. The thyroid gland is composed of follicles and stromal tissues.
B. Exophthalmic goitre is a form of hypothyroidism.
C. Parathyroid hormone is a hypercalcemic hormone.
D. Thyrocalcitonin is a steroid hormone.
E. Cortisol is a glucocorticoid that stimulates gluconeogenesis, lipolysis, and proteolysis.
(1) A, C and E
(2) A, B and C
(3) B, D and E
(4) C, D and E

143. Which of the following techniques relies on the principle of antigen-antibody interaction to detect a pathogen?
(1) Polymerase Chain Reaction (PCR)
(2) Autoradiography using a radioactive probe
(3) Enzyme Linked Immuno-sorbent Assay
(4) Gel Electrophoresis

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Alkaloids	(I)	Concanavalin A
(B)	Toxins	(II)	Rubber, gums, cellulose
(C)	Lectins	(III)	Morphine, Codeine
(D)	Polymeric substances	(IV)	Abrin, Ricin

Choose the **correct** answer from the options given below:
(1) A-III, B-IV, C-II, D-I
(2) A-IV, B-III, C-I, D-II
(3) A-III, B-IV, C-I, D-II
(4) A-II, B-IV, C-III, D-I

145. Given below are two statements:
Statement I: A sparged stirred-tank bioreactor is designed to allow sterile air bubbles to pass through, significantly increasing the oxygen transfer area.
Statement II: Downstream processing involves only the separation of the product.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. The presence of which two specific substances in the urine is highly indicative of Diabetes mellitus?
(1) Uric acid and high urea
(2) Glucose and ketone bodies
(3) Amino acids and bilirubin
(4) Red blood cells and excess proteins

147. At what stage of pregnancy, the body of the foetus first observed to be covered with fine hair, alongside the separation of eye-lids and formation of eyelashes?
(1) By the end of the first month
(2) By the end of the 12th week
(3) During the fifth month
(4) By the end of about 24 weeks

148. How many eggs are typically released by a human ovary in a month, and how does this change in the case of twins?
(1) Normally 1 egg; 2 eggs if identical twins; 2 eggs if fraternal twins
(2) Normally 1 egg; 1 egg if identical twins; 2 eggs if fraternal twins
(3) Normally 2 eggs; 2 eggs if identical twins; 1 egg if fraternal twins
(4) Normally 1 egg; 2 eggs if identical twins; 1 egg if fraternal twins

149. What is the optimal temperature used for the heat shock step in making a bacterial cell competent for transformation?
(1) 37°C
(2) 42°C
(3) 72°C
(4) 94°C

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The theory of spontaneous generation was ultimately dismissed.
Reason R: Louis Pasteur demonstrated through careful experimentation with pre-sterilized swan-neck flasks that life only comes from pre-existing life.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. Which of the following phenomenon **correctly** describes metastasis?
(1) The phenomenon where cells stop dividing upon physical contact with surrounding cells.
(2) The activation of proto-oncogenes leading to localized benign tumors.
(3) The process where sloughed-off tumor cells reach distant sites through the blood and start new tumors there.
(4) The induction of DNA damage by physical agents like X-rays causing a primary lesion.

152. According to the sliding filament theory of muscle contraction, what happens to the structural bands when a skeletal muscle fibre maximally contracts?
(1) Both the 'A' band and 'I' band get reduced in length.
(2) The 'H' zone disappears, the 'I' band shortens, and the 'A' band retains its length.
(3) The 'A' band shortens, while the 'Z' lines are pushed outwards.
(4) The thick filaments contract in length causing the 'A' band to disappear.

[Contd...

153. Consider the following statements regarding disorders of the human circulatory system.

A. Hypertension refers to blood pressure higher than the normal baseline of 120/80 mm Hg, where 120 represents diastolic pressure.
B. Atherosclerosis, causes the lumen of arteries to become narrower.
C. Angina pectoris is strictly confined to the elderly population.
D. Heart failure is the condition where the heart is not pumping blood effectively.
E. A heart attack means the heart stops beating.

Choose the **correct** statements.

(1) B and D only (2) A, B, and D
(3) B, C, and E (4) A, C, and E

154. How is the majority of CO_2 transported from the tissues to the alveoli?

(1) As carboxyhemoglobin.
(2) Dissolved directly in the blood plasma.
(3) As bicarbonate ions
(4) As carbamino-hemoglobin.

155. What is the function of the Bidder's canal in frogs?

(1) It stores bile.
(2) It acts as a passage for sperms traveling from the vasa efferentia through the kidney to the urinogenital duct.
(3) It forms a specialized venous connection between the liver and the intestine.
(4) It serves as a specialized chamber for the storage of unfertilized ova in the female.

156. Identify the **correct** statements regarding central nervous system of humans.

A. The cerebral cortex is the white matter.
B. The limbic system, along with the hypothalamus, is involved in the regulation of sexual behaviour.
C. The midbrain contains four round swellings
D. The cerebellum has a very convoluted surface.
E. The medulla oblongata is not connected to the spinal cord.

(1) A, B, and D (2) B, C, and D
(3) C, D, and E (4) A, C, and E

157. Hormone of which gland is responsible for maintaining the sleep-wake cycle?

(1) Pituitary gland
(2) Hypothalamus
(3) Pineal gland
(4) Thymus gland

158. Given below are two statements:

Statement I: When ready-made antibodies are directly given to protect the body against foreign agents, it is called passive immunity.

Statement II: The use of a thermostable DNA polymerase from *Thermus aquaticus* allows the PCR to continue through high-temperature denaturation steps.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. Which of the following respiratory capacities **cannot** be measured by spirometer?

(1) IC
(2) EC
(3) FRC
(4) VC

160. Given below are two statements:

Statement I: Humulin does not induce unwanted immunological responses in diabetic patients.

Statement II: Humans are able to maintain a constant body temperature regardless of the environmental temperature.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect.
(2) Both Statement I and Statement II are correct.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

161. HIV infection primarily targets and reduces which type of cells, causing immunodeficiency?

(1) B-lymphocytes
(2) Natural killer cells
(3) Helper T-lymphocytes
(4) Erythrocytes

[Contd...

162. Given below are two statements:
Statement I: In blue-white screening, if the gene of interest is inserted into the coding sequence of beta-galactosidase, as a result, recombinant colonies will be blue.
Statement II: White muscle fibres possess very few mitochondria.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

163. Which of the following **correctly** describes the effect of parasympathetic neural signals on heart?
(1) Increase the rate of the heartbeat
(2) Decrease the rate of the heartbeat
(3) Increase cardiac output
(4) Maximise stroke volume

164. Match **List-I** with **List-II**.

List-I (Application of transgenic animals)		**List-II (Specific example/description)**	
(A)	Understanding normal physiology and development of cells	(I)	Using mice to test batches of the polio vaccine
(B)	Study of disease	(II)	Producing human protein alpha-1-antitrypsin for emphysema
(C)	Biological products	(III)	Serving as models for Alzheimer's and Cystic fibrosis
(D)	Vaccine safety	(IV)	Studying complex factors like insulin-like growth factor

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-II, D-IV
(2) A-III, B-IV, C-I, D-II
(3) A-IV, B-II, C-III, D-I
(4) A-IV, B-III, C-II, D-I

165. Which of the following is inserted into the female reproductive tract to cover the cervix during coitus?
(1) Progestasert (2) Vaults
(3) LNG-20 (4) Multiload 375

166. In an enzyme-catalyzed metabolic reaction, the role of the enzyme is to:
(1) Increase the potential energy of the substrate.
(2) Lower the energy level of product.
(3) Bring down the activation energy barrier
(4) Create an unstable product

167. A patient exhibits a rapid spasm in the muscles due to a severely low concentration of Ca^{2+} in the body fluid. This specific muscular disorder is termed:
(1) Myasthenia gravis
(2) Muscular dystrophy
(3) Osteoporosis
(4) Tetany

168. *Agrobacterium* vectors introduce nematode-specific genes into tobacco plants. What is the immediate result in the plant cells?
(1) Produces toxic insecticidal protein
(2) Produces antisense RNA degrading nematode genome
(3) Produces sense and antisense RNA forming dsRNA
(4) Produces mutant allele replacing nematode gene

169. When a person goes up a hill, what happens to their respiratory process?
(1) Breathing rate decreases
(2) Breathing rate increases
(3) Oxygen consumption stops temporarily
(4) Carbon dioxide is no longer produced

170. Identify the **incorrectly** matched pair:
(1) Ball and socket joint - Between humerus and pectoral girdle
(2) Hinge joint - Knee joint
(3) Pivot joint - Between the carpals
(4) Saddle joint - Between carpal and metacarpal of thumb

171. Match **List-I** with **List-**II.

List-I		**List-II**	
(A)	Gastrin	(I)	Part of limbic system
(B)	Amygdala	(II)	Middle layer of uterus
(C)	ZIFT	(III)	ART
(D)	Myometrium	(IV)	Secreted by GIT

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-III, D-II
(2) A-I, B-IV, C-II, D-III
(3) A-IV, B-I, C-II, D-III
(4) A-II, B-III, C-IV, D-I

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172. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The male gametes are formed within the interstitial spaces.
Reason R: Leydig cells secrete androgens, which stimulate spermatogenesis.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is not the correct explanation of A.

173. Which gas mixture and temperature are maintained in the flask in Miller's experiment?
(1) CH_4, H_2, NH_3 and water vapour at 800°C
(2) CO_2, H_2, NH_3 and water vapour at 600°C
(3) CH_4, O_2, NH_3 and water vapour at 800°C
(4) CH_4, H_2, N_2 and water vapour at 1000°C

174. Which of the following fructose polymer serves as a polymeric reserve in plants?
(1) Cellulose (2) Glycogen
(3) Starch (4) Inulin

175. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Sporozoites	(I)	Hyperglycemic hormone
(B)	Pons	(II)	Consists of fibre tracts that interconnect different regions of brain
(C)	Glucagon	(III)	Mineralocorticoid
(D)	Aldosterone	(IV)	Infective stage of *Plasmodium* for humans

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-II, B-IV, C-III, D-I
(3) A-IV, B-II, C-I, D-III
(4) A-III, B-II, C-IV, D-I

176. Which STI is **not** completely curable even if detected early?
(1) Syphilis (2) Chlamydiasis
(3) Genital herpes (4) Gonorrhoea

177. According to the Hardy-Weinberg equilibrium, if the frequency of a recessive allele in a randomly mating population is 0.4, what will be the expected frequency of the heterozygous individuals in the next generation?
(1) 0.16 (2) 0.36
(3) 0.48 (4) 0.84

178. Which protein is the most abundant in the animal world?
(1) RuBisCO
(2) Haemoglobin
(3) Collagen
(4) Insulin

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The thorns of *Bougainvillea* and the tendrils of *Cucurbita* represent homology.
Reason R: They are anatomically similar structures adaptated to different needs, indicating divergent evolution.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

180. Which of the following indicates sexual dimorphism in Aschelminthes?
(1) Males and females are not distinct.
(2) Females are often longer than males.
(3) Males are always longer than females.
(4) They are hermaphrodites.

■■■

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